

PART 1 -- GENERAL

1.1 SCOPE: The work of this section includes providing and installing buried pipe, and fittings.

1.2 REFERENCES:

- A. American Society for Testing and Materials (ASTM)
- B. American National Standards Institute (ANSI)
- C. American Water Works Association (AWWA)
- D. Federal Specifications (FS)

1.3 SUBMITTALS: In accordance with Section 01 33 00, submit:

- A. Certificate of compliance.
- B. Manufacturer's literature and product data sufficient to demonstrate compliance with the specified requirements.
- C. Manufacturer's installation instructions and recommendations.
- D. Test results.

PART 2 -- PRODUCTS

2.1 GENERAL: Pipe shall have joints as called for in the specifications or indicated on the Drawings. All pipeline and appurtenant materials that will be in contact with potable water shall be NSF 61 rated for potable water service.

2.2 HARDWARE MATERIALS: Hardware used for the assembly of piping systems, flanges, joints, and appurtenances (including coupling bolts, tie rods, mechanical restraint systems, and the like) shall comply with the following:

- A. Interior dry locations, or exterior above grade: Hot dip galvanized steel.
- B. Moist locations (any interior or exterior space wholly or partially below grade level including vaults or pits, or having a wall or ceiling forming head space of part of a clean water channel or basin): Type 304 Stainless Steel with "Never Gall" factory applied coating system.
- C. Submerged locations and in corrosive areas (corrosive meaning spaces with NEC electrical classifications of Class 1 Divisions 1 and 2, in chemical storage and pumping areas, and in head space of channels or basins containing process liquids): Type 316 stainless steel with "Never Gall" factory applied coating system.
- D. Buried in earth: Type 304 stainless steel with "Never Gall" factory applied coating system.

2.3 DUCTILE IRON PIPE, FITTINGS, AND APPURTENANCES:

- A. Pipe: Ductile-iron pipe, conforming to AWWA C151/ANSI A21.51. Provide Special Thickness Class 52, unless specified otherwise.
- B. Joints: Ductile iron pipe shall be flanged, push-on, or mechanical joint as shown on the Drawings. In general, flanged pipe shall be used above ground or where exposed in vaults etc., while push-on or mechanical joint shall be used where buried in earth.
 - 1. Mechanical and Push-On: In accordance with AWWA C111/ANSI A21.11, and the gaskets shall be of the materials identified in the pipe schedule and as provided by the pipe manufacturer.
 - 2. Flanged joints: In accordance with AWWA C115/ANSI A21.15 and ANSI B16.1 Class 125, except provide Class 250 when specifically shown on the Drawings or required for connection to appurtenances or valves. Flanges shall be accurately shop faced and drilled true. Where tap or stud bolts are shown or otherwise required, flanges shall be tapped. Flanged pipe joints shall be made with full face gaskets of cloth inserted rubber compound, not less than 1/8-inch in thickness. Rubber compound as listed in the pipe schedule. Gaskets shall be pressure-rated for at least 1.5 times the specified test pressure of the pipeline, submit the pressure rating for review.
- C. Fittings: Ductile-iron conforming to the requirements set forth in AWWA C110/ANSI A21.10 or AWWA C153/ANSI A21.53. Provide Class 250 minimum. Joint type shall be as specified, and as shown on the Drawings or appropriate for the installation location. All mechanical joint solid sleeves shall be long pattern.
- D. Spools and Wall Pipe: Spools shall be cast as fittings in accordance with AWWA C110 or fabricated from Special Thickness Class 53 ductile iron pipe in accordance with AWWA C115. Wall pipe shall have collars integrally cast. Collars shall be located so as to be in the center of the concrete wall or floor into which they are to be placed.
- E. Interior Lining: Pipe, spools and fittings shall be either cement mortar lined and seal coated in accordance with AWWA C104/ANSI A21.4, or Protecto 401 ceramic epoxy lined as indicated in the pipe schedule.
- F. Exterior Coating:
 - 1. Pipe, spools and fittings to be buried in earth or installed within below-grade vaults shall be furnished with standard thickness asphalt coating per AWWA C151.
 - 2. Pipe, spools and fittings to be installed above ground shall be furnished with a factory prime coat, for finish coating in the field with the appropriate system specified in Specification 09 90 00.
 - 3. Pipe, spools and fittings to be installed in submerged locations shall be supplied by the factory bare, for shop blasting and application of the specified submerged protective coating system. Asphalt coated or prime coated pipe shall not be used in exposed or submerged locations.

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G. Polyethylene Encasement: Ductile-iron pipe, spools and fittings buried in earth shall be encased in polyethylene film in accordance with the following requirements of AWWA C105/ANSI A21.5.

1. Polyethylene: The polyethylene film shall be manufactured of virgin polyethylene material conforming to the requirements of ASTM D-1248. The raw materials shall be Type 1, Class A (natural) or C (Black), Grade E-1 with flow rate of 0.4 maximum and dielectric strength of 10^{15} ohm-cm³ minimum.
2. Polyethylene Film: The finished polyethylene film shall have a minimum nominal thickness of 0.008-inch (8 mil), and the minus tolerance on thickness shall not exceed 10 percent of the nominal thickness. The film shall have a minimum tensile strength of 1200 psi with an elongation of 300 percent minimum. The dielectric strength shall be 800 volts/mil thickness minimum.
3. Tube Size or Sheet Width: The tube size or sheet width for each pipe diameter shall be as listed in the following table:

Nominal Pipe Diameter (inch)	Minimum Width (inch)	
	Flat Tube	Sheet
4	16	32
6	20	40
8	24	48
10	27	54
12	30	60
14	34	68
16	37	74
18	41	82
20	45	90
24	54	108
30	67	134
36	81	162
42	95	190
48	108	216
54	121	242

H. Thrust Restraint: All pressurized ductile iron piping systems shall be fully restrained against thrust. Thrust restraint shall be achieved by the use of both concrete thrust blocks and mechanical restraints.

Thrust blocks shall be installed in accordance with the details shown on the Drawings. Concrete shall be Class A, Type II, and have a minimum 28-day compressive strength of 3,000 psi. Concrete shall be placed against polywrapped fittings – never against bare fittings.

Mechanical restraints shall be installed to the minimum restrained lengths shown on the details on the Drawings, for horizontal and vertical bends, other fittings, valves and dead-ends. Restraints shall conform to the following:

1. Push-On Joint Restraints:

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- a. Restraint ring with serrated wedges on the spigot end of the pipe, with a split bell ring that engages behind the bell without serrations on the other. Ductile iron construction per ASTM A536 with protective coating, hardware material as specified elsewhere herein. EBBA Iron Megalug Series 1700.
 - b. As an alternative, proprietary restrained pipe joints may be provided for both pipe and fittings; TR FLEX as manufactured by U.S. Pipe and meeting the requirements for ductile iron pipe and fittings specified elsewhere herein. Restrained push-on joint pipe and fittings shall be capable of being deflected after assembly. One set of each special assembly tool recommended by the manufacturer shall be supplied to the Owner.
2. Mechanical Joint Restraints: Restraint ring with serrated wedges and incorporating a follower gland, designed to bolt to a mechanical joint drilling pattern. Ductile iron construction per ASTM A536 with protective coating, hardware material as specified elsewhere herein. EBBA Iron Megalug Series 1100.
 3. Tie Rods: Where specifically designated on the Drawings, details, or Special Construction Provisions, fitting restraints shall be made with tie rods and pipe clamps or special fitting bolts. Tie rod restraint systems shall have a minimum of 2 bolts or rods per joint or clamp. Minimum tie bolt and rod sizes are as follows:

<u>Pipe Size (inch)</u>	<u>Bolt and Rod Diameter and Quantity (inch)</u>
4 - 8	5/8 x 2
10 - 12	3/4 x 2
14	7/8 x 2
16	1 x 2
18 - 20	1-1/4 x 2
24	1-1/2 x 4

Tie bolts and rods shall be fabricated from materials specified elsewhere herein. Retainer clamps shall be equal to "socket clamp," Figure 595, as manufactured by ITT-Grinnell.

2.4 POLYVINYL CHLORIDE (PVC) PIPE (PRESSURE):

- A. Pipe shall conform to AWWA C900-16 "Polyvinyl Chloride (PVC) Pressure Pipe 4 inch through 60 inch". The pipe material shall be of virgin source, conform to ASTM D1784 Cell Classification 12454, and be integrally pigmented purple in color.

Provide Pressure Class 235, DR18, unless specified otherwise in Section 00 80 00 or on the Drawings.

Provisions shall be made for contraction and expansion at each joint with a rubber ring, and integral thickened bell as part of each joint. Integral joints shall conform to ASTM D3139. Gaskets shall conform to ASTM F477. Pipe shall be supplied in nominal laying lengths of 20 feet. Pipe and fittings shall be assembled with a non-toxic lubricant. Each length of pipe shall have marked on the exterior the following:

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Manufacturer's Name or Trademark

Nominal Pipe Size/Dimension Ratio

PVC Cell Classification (e.g. 12454-B)

Legend – Type AWWA C900 Pressure Pipe

- B. Fittings: Ductile iron with mechanical joints as specified elsewhere herein. Provide polywrapping in accordance with AWWA C105.
- C. Thrust Restraint: Pressurized PVC piping systems shall be fully restrained against thrust. Thrust restraint shall be achieved by the use of both concrete thrust blocking and mechanical restraints.

Thrust blocks shall be installed in accordance with the details shown on the Drawings. Concrete shall be Class A, Type II, and have a minimum 28-day compressive strength of 3,000 psi. Concrete shall be placed against polywrapped fittings – never against bare fittings.

Mechanical restraints shall be installed to the minimum restrained lengths shown on the Drawings, for horizontal and vertical bends, other fittings, valves and dead-ends. Mechanical restraints shall be polywrapped in accordance with AWWA C105. Restraints shall conform to the following:

1. Straight Pipe: Split ring, with continuous serrations around the full circumference on the spigot end of the joint, and a non-serrated ring that rests against the back of the bell on the opposite end of the joint. Ductile iron construction per ASTM A536 with protective coating, hardware material as specified elsewhere herein. EBBA Iron Series 1600 or Series 2800 for C900 pipe.
2. Mechanical Joint Fittings: Restraint ring with serrated wedges and incorporating a follower gland, designed to bolt to a mechanical joint drilling pattern. Ductile iron construction per ASTM A536 with FBE protective coating and lining, hardware material as specified elsewhere herein. EBBA Iron Series 2000PV for C900 pipe.
3. Tie Rods: Where specifically designated on the Drawings, details, or Special Construction Provisions, fitting restraints shall be made with tie rods and pipe clamps or special fitting bolts. Tie rod restraint systems shall have a minimum of 2 bolts or rods per joint or clamp. Minimum tie bolt and rod sizes are as follows:

<u>Pipe Size (inch)</u>	<u>Bolt and Rod Diameter and Quantity (inch)</u>
4 - 8	5/8 x 2
10 - 12	3/4 x 2
14	7/8 x 2
16	1 x 2
18 - 20	1-1/4 x 2
24	1-1/2 x 4
36	1-5/8 x 4

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Tie bolts and rods shall be fabricated from materials specified elsewhere herein. Retainer clamps shall be equal to “socket clamp,” Figure 595, as manufactured by ITT-Grinnell.

2.5 POLYVINYL CHLORIDE (PVC) PIPE (GRAVITY SEWER SERVICE)

- A. 4 to 15-inch diameter: polyvinyl chloride pipe (PVC) suitable for gravity sewer service. The pipe material shall be made of PVC plastic having a cell classification of 12454-B or 12364-C or 12454-C or 13364-B (with a minimum tensile modules of 500,000 psi) as defined in ASTM D1784. PVC pipe and fittings shall meet or exceed the material requirements of ASTM D3034 and thickness requirements of SDR-35 (4-inch to 15-inch diameter). The pipe material shall be integrally pigmented green in color.
- B. Provisions must be made for contraction and expansion at each joint with a rubber ring and integral thickened bell as part of each joint. Gaskets shall conform to ASTM F477. Pipe shall be supplied in nominal laying lengths of 20 feet. Pipe and fittings shall be assembled with a non-toxic lubricant. Each length of pipe and all fittings shall have marked on the exterior the following:

Manufacturer's Name or Trademark

Nominal Pipe Size

PVC Cell Classification (e.g. 12454-B)

Legend - Type PSM SDR-35 Sewer Pipe

ASTM-D3034

- C. Fittings to be used with the PVC pipe (e.g. tee or wye fittings for service taps, etc) shall be those manufactured by the manufacturer of the pipe and shall match the specification and classification of the pipe to which they are connected. Special fabricated fittings shall be a completely manufactured unit with either bells or spigots on each connection that are an exact duplication of the bells and spigots on the pipeline. Fittings with any other type of connections will not be accepted.
- D. Sewer service lines shall connect to the sewer main at either a tee or wye.
- E. Saddles, for use with connection to existing mains, shall be manufactured of the same materials as the sewer pipeline or of PVC. Saddles shall provide for a right-angle entry into the sewer main. The saddle shall be contoured to fit the sewer main pipe and shall be permanently attached thereto with an epoxy glue or stainless steel bands. Saddles shall be specifically constructed to connect the service line to the main line without modification of either.

2.6 HIGH DENSITY POLYETHYLENE (HDPE) – DIPS PIPE: Shall be high-density polyethylene (PE 4710 resin, ductile iron pipe size O.D) pressure pipe primarily intended for the transportation of water and sewage either buried or above grade. References include: AWWA C901 Polyethylene (PE) pressure Pipe & Tubing, ½ inch through 3 inch for water, AWWA C906 Polyethylene (PE) pressure Pipe & Fittings, 4 inch through 63 inch, 1.2.3 ASTM D3035 Standard Spec for PE Pipe (DR-PR) Based on Controlled Outside Diameter, ASTM D3261 Butt Heat Fusion

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PE Fittings for PE Pipe & Tubing, ASTM D3350 Standard Specification for PE Pipe & Fittings Materials, ASTM D1238 Melt Flow Index, ASTM D1505 Density of Plastics, ASTM D2837 Hydrostatic Design Basis, TR-33/2005 Generic Butt Fusion Joining Procedure for Field Joining of PE Pipe.

- A. Submit results of the following tests for resin identification: Melt Flow Index ASTM D1238 ASG Housing Distribution Lines Replacement Project 2 and Density ASTM D1505
- B. Manufacturer: HDPE pipe and fittings shall be from a single manufacturer, who is fully experienced, reputable, and qualified in the manufacture of the HDPE pipe. Qualified manufacturers shall be: PLEXCO Division of Chevron Chemical Company, DRISCOPIPE as manufactured by Phillips Products Co., Inc., SCLAIRPIPE.
- C. Contractor shall check each length of pipe produced for the items listed below. The results of all measurements shall be recorded on production sheets, which become part of the manufacturer's permanent records. Pipe in process shall be checked visually, inside and out for cosmetic defects (grooves, pits, hollows, etc.) Pipe outside diameter shall be measured using a suitable periphery tape to ensure conformance with ASTM F714 or ASTM D3035, whichever is applicable.

Pipe wall thickness shall be measured at 12 equally spaced locations around the circumference at both ends of the pipe to ensure conformance with ASTM F714 or ASTM D3035, whichever is applicable. Pipe length shall be measured. Pipe marking shall be examined and checked for accuracy. Pipe ends shall be checked to ensure they are cut square and clean. Subject inside surface to a "reverse bend test" to ensure the pipe is free of oxidation (brittleness).

- D. Stress Regression Testing: The polyethylene pipe manufacturer shall provide certification that stress regression testing has been performed on the specific polyethylene resin being utilized in the manufacture of this product. This stress regression testing shall have been done in accordance with ASTM D2837 and the manufacturer shall provide a product supplying a minimum Hydrostatic Design Basis (HDB) of 1,600 psi as determined in accordance with ASTM D2837.
- E. Compatibility: Contractor is responsible for compatibility between pipe materials, fittings, and appurtenances.
- F. Materials for Pipe size 4-inch diameter and larger: Materials used for the manufacture of polyethylene pipe and fittings shall be made from a PE4710 high density polyethylene resin compound. Dimensions and workmanship shall be as specified by ASTM F714. HDPE fittings and transitions shall meet ASTM D3261. HDPE pipe shall have a minimum density of 0.955 grams per cubic centimeter. All HDPE pipe and fittings shall have a Hydrostatic Design Basis (HDB) of minimum 1,600 psi.
- G. Fittings: Molded fittings and fabricated fittings shall be fully pressure rated to match the pipe SDR pressure rating to which they are made. Fittings shall be molded or fabricated by the manufacturer. No Contractor-fabricated fittings shall be used. The manufacturer of the HDPE pipe shall supply HDPE fittings and accessories as well as any adapters and/or specials required to perform the work. Fittings shall be installed using butt-fused fittings, thermo-fused fittings/couplings, or flanged adapters. No size-on-size wet taps shall be permitted.

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All transition from HDPE pipe to ductile iron or PVC shall be made per the HDPE pipe manufacturer's recommendations and specifications. A molded flange connector adapter within a carbon steel back-up ring assembly shall be used for pipe type transitions. Ductile iron back-up rings shall mate with cast iron flanges per ANSI B16.1. A 316 stainless steel back-up ring shall mate with a 316 stainless steel flange per ANSI. No solid sleeves shall be allowed between such material transitions.

- H. Pipe Identification: The following shall be continuously indent printed on the pipe or spaced at intervals not exceeding 5-feet: Name and/or trademark of the pipe manufacturer, Nominal pipe size, Dimension ratio, The letters PE followed by the polyethylene grade in accordance with ASTM, followed by the hydrostatic design basis in 160's of psi, e.g., PE 4710. Color Identification, either stripped by co-extruding longitudinal identifiable color markings or shall be solid in color and as follows: Green – Sanitary Sewer.
- I. Joining Method: The pipe shall be joined with butt, heat fusion joints as outlined in ASTM D2657 and conform to the Generic Butt Fusion Joining Procedure for Field Joining of Polyethylene Pipe, Technical Report TR-33/2005, published by the Plastic Pipe Institute (PPI). All joints shall be made in strict compliance with the manufacturer's recommendations. A factory qualified joining technician as designated by pipe manufacturer or experienced, trained technician shall perform all heat fusion joints. Lengths of pipe shall be assembled into suitable installation lengths by the butt fusion process. All pipes so joined shall be made from the same class and type of raw material made by the same raw material supplier. Pipe shall be furnished in standard laying lengths not to exceed 50 feet and no shorter than 20 feet.

On days butt fusions are to be made, the first fusion shall be a trial fusion. The following shall apply: Heating plate surfaces shall be inspected for cuts and scrapes and shall be free of dirt and residue. Heater surfaces shall be between 400°F (minimum) to 450°F (maximum). Measure the temperature @ 12:00, 3:00, 6:00 and 9:00 o'clock positions using a pyrometer or infrared thermometer at locations where the heating plate will contact the pipe/fitting ends. The maximum temperature difference between any two points on a single heating surface must not exceed 24°F. If this temperature is exceeded, the heating plate shall be cleaned per the manufacturer's recommendations. The fusion or test section shall be cut out after cooling completely for inspection. The test section shall be 12" or 30 times (minimum) the wall thickness in length and 1" or 1.5 times the wall thickness in width (minimum).

The joint shall be visually inspected as to continuity of "beads" from the melted material, and for assurance of "cold joint" prevention (i.e. – joint shall have visible molded material between walls of pipe). Joint spacing between the walls of the two ends shall be a minimum of 1/16" to a maximum 3/16". The polyethylene flange adapters at pipe material transitions shall be backed up by stainless steel flanges conforming to ANSI B16.1 and shaped as necessary to suit the outside dimensions of the pipe. The flange adapter assemblies shall be connected with corrosion resisting bolts and nuts of Type 316 Stainless Steel as specified in ASTM A726 and ASTM A307. All bolts shall be tightened to the manufacturer's specified torques. Bolts shall be tightened alternatively and evenly. After installation, apply a bitumastic coating to bolts and nuts.

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2.7 WATER SERVICES 2-INCHES AND SMALLER: All components shall be suitable for potable water use, and manufactured in accordance with AWWA C800 using lead-free copper alloy UNS No. C89520 or copper alloy UNS No. C89833, in accordance with ASTM B584, as applicable. Components shall be certified to comply with NSF 61, NSF 61 Annex G, NSF 372.

- A. Service Line 2" and smaller: Type K Copper. Seamless copper water tube. ASTM B88, UNS No. 12200. Furnished in coils and annealed.
- B. Tapping Saddles:
 - 1. For use with Ductile or Cast Iron Pipe: Double bronze straps. AWWA standard taper threads. Mueller BR2B series, Ford 202B-NL series.
 - 2. For use with PVC Pipe: Bronze saddles shall provide full support around the circumference of the pipe, and have a bearing area of sufficient width along the axis of the pipe so that the pipe will not be distorted when the saddle is tightened. Saddles shall be double strap, bronze. Mueller H-13000 or H-134000 series, Ford S90-NL series.
- C. Corporation Stops: AWWA Standard taper thread by copper flare, unless other end connections are specified on the Drawings. Mueller B25000N, Ford FB-600-NL series.
- D. Insulators: Provide electrical insulators at all corporation stops for services on ductile or cast iron mains. Mueller, Ford.
- E. Curb Stops: Flare by flare end connections, unless other end connections are specified on the Drawings. Mueller B25204N, Ford B22-NL series. Provide a cast iron box for each curb stop. Star Pipe Products SB90ES series, or RWB 145RHD series where located in roadways.
- F. Couplings and fittings: Flare by flare end connections, unless other end connections are specified on the Drawings. Ford C28-NL series.

2.8 BURIED UTILITY WARNING TAPE: Metallic detectable type, three (3) inch wide plastic metallic type consisting of a color-coded polyethylene or melinex film, a solid core aluminum detection layer and other layers recommended by the manufacturer to provide durability and longevity. The tape and ink shall be resistant to acids, alkalis and other components likely to be encountered in soils. Tape shall be color-coded blue, with repeating text "Caution Buried Pipe Below" or equivalent. Sentry Line Detectable Terra Tape by Reef Industries; Detectable Marking Tape by T. Christy Enterprises.

2.9 INSULATION BOARD: Insulation where shown on the Drawings shall be Type SM rigid blue closed cell foam or Type HI-40 as manufactured by the DOW Chemical Corporation. Type HI-40 shall be used beneath foundations, all traveled way-in roads and streets, highway shoulders and within 15 feet of pavement unless otherwise specified on the Drawings.

PART 3 -- EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

- A. Do not lay pipe when trenches or weather conditions are unsuitable for such work.

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- B. Each pipe length and fitting interior, interior surface of bells, and exterior surface of spigots shall be cleaned of all foreign material before placing it in the trench and shall be kept clean all times thereafter. Each item shall also be examined for cracks and other defects before installation.
- C. Field cutting of pipe for inserting valves, fittings, or closure pieces shall be done in a neat and workmanlike manner without damage to the pipe, and so as to leave a smooth end at right angles to the axis of the pipe.
- D. Each pipe length shall be laid true to line and grade, without intermediate high or low points not shown on the Drawings.
- E. Pipe shall be laid in a dry (dewatered) trench and shall not be used for draining water from the trench.
- F. Whenever the pipe is left unattended or pipe laying is not in progress, temporary plugs shall be installed at all openings. Temporary plugs shall be watertight and of such design as to prevent debris and animals from entering the pipe.
- G. In some special circumstances it may be necessary to install the pipeline shallower than the minimum required depth of cover, such as to avoid other utilities or to achieve a specific slope or grade. When the depth of cover is less than 3.5-ft, install the specified insulation board directly on the top of pipe zone bedding over the pipeline, extending 2-ft on each side of the pipe centerline, and continuous for the entire length of the reduced depth of cover. Tape seams of abutting insulation boards together.
- H. Contractor shall install the materials in accordance with the manufacturer's recommendations. If there is a conflict between the Contract Documents and the manufacturer's instructions, the Contractor shall obtain resolution from the Engineer before proceeding with the work.
- I. Pipe, fittings, and other accessories shall be loaded and unloaded by lifting with hoists or skidding so as to avoid shock or damage to them. Under no circumstances shall any materials be dropped. Pipe handled on skidways shall not be skidded or rolled against pipe already on the ground. Skidding which damages protective coatings will not be permitted.

In distributing the material at the site of the work, each piece shall be unloaded opposite or near the place where it is to be laid in the trench to minimize moving more than once.

All pipe and fittings shall be so handled that the coating and lining will not be damaged. If, however, any part of the coating or lining is damaged, the repair shall be by the Contractor at their expense in a manner satisfactory to the Engineer. Any area damaged beyond repair must be cut off and discarded.

- J. Pipe and fittings will be checked for out-of-round or damaged joints, interior and exterior surface damage, gasket damage, and other requirements listed herein. Pipe or appurtenant material found defective will be rejected. Any material rejected at the job site shall be marked "Rejected", and the Contractor shall remove it immediately from the job site.

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- K. Do not store materials directly on the ground. Polyvinyl chloride (PVC) pipe shall be stored under opaque cover that does not transmit UV light.

3.2 INSTALLATION OF DUCTILE-IRON PIPELINES: Installation of pipe shall conform to the recommendations contained in “A Guide for Installation of Ductile-Iron Pipe,” published by the Ductile Iron Pipe Research Association. A copy shall be available at the job site.

- A. Pipe Laying: Pipe shall be laid with bell ends facing in the direction of laying. Pipe shall be laid on the bedding with support over the full length of the pipe barrel.
- B. Pipe joint deflections shall not exceed the amount shown in the following table.

DUCTILE-IRON PIPE DEFLECTION

Size of Pipe (inches)	Bend in One Joint (degrees)	Deflection in One 18-Foot Length (inches)	Approximate Radius of Curve Produced by Succession of 18-Foot Joints (feet)
4	4	15	250
6	4	15	250
8	4	15	250
10	4	15	250
12	4	15	250
14	2.5	9	450
16	2.5	9	450
18	2.5	9	450
20	2.5	9	450
24	2.5	9	450

The information in the columns referring to the deflection and the approximate radii shall be adjusted for pipe lengths different than 18-foot lengths. To lay pipelines on curved alignment with shorter radius if called for on the Drawing, the Contractor use shorter pipe lengths. Double hubs may be used to lay pipelines on curved alignment.

Vertical deflections shall not exceed any of the referenced values.

- C. The cutting of pipe for inserting valves, fittings, or closure pieces shall be done in a neat and workmanlike manner without damage to the pipe or cement lining so as to leave a smooth end at right angles to the axis of the pipe. The flame cutting of pipe by means of an oxyacetylene torch will not be allowed. The pipe end shall be beveled and free of sharp edges that could damage the gasket during installation.
- D. Jointing of Mechanical Joints: The last 8 inches of the pipe spigot and the inside of the bell of the mechanical joint shall be thoroughly cleaned to remove oil, grit, tar (other than standard coating), and other foreign matter from the joint, and then painted with a manufacturer supplied lubricant or soap solution made by dissolving one-half cup of granulated soap in one gallon of water. The cast-iron gland shall then be slipped on the spigot end of the pipe with the lip extension of the gland toward the spigot end. The gasket shall be painted with the lubricant or soap solution and placed on the spigot end of the pipe to be laid, with the thick edge toward the gland.

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The entire section of the pipe being laid shall be pushed forward to seat in the spigot end of the bell of the pipe in place. The gasket shall then be pressed into place within the bell, being careful to have the gasket evenly located around the entire joint. The cast-iron gland shall be moved along the pipe into position for bolting, all of the bolts inserted, and the nuts screwed up tightly with fingers. All nuts shall then be tightened with a suitable (preferably torque-limiting) wrench. The torque for various sizes of bolts shall be as follows:

<u>Size (Inches)</u>	<u>Range of Torque</u> <u>ft. - lb.</u>
$\frac{5}{8}$	45 - 60
$\frac{3}{4}$	75 - 90
1	100 - 120
$1\frac{1}{4}$	120 - 150

Nuts spaced 180 degrees apart shall be tightened alternately in order to produce an equal pressure on all parts of the gland.

- E. Jointing of Push-On Joints: In jointing the pipe, the exterior 4 inches of the pipe at the spigot end and the inside of the adjoining bell and particularly the groove for the gasket shall be thoroughly cleaned to remove oil, grit, tar (other than standard coating), and other foreign matter. The proper gasket supplied with the pipe shall be placed in the bell as described by the pipe manufacturer so it will spring into its proper place inside the pipe bell. A thin film of the pipe manufacturer's joint lubricant shall be applied to the gasket over its entire exposed surface. The spigot end of the pipe shall then be wiped clean and inserted into the bell to contact the gasket. Then the pipe shall be forced all the way into the bell by crowbar or by jack and choker slings. The location of the gasket shall be checked with a gauge or tool designed for that purpose to assure that the gasket is in the proper position.
- F. Installation of Polyethylene Encasement: When required, the polyethylene encasement shall prevent contact between the pipe and the surrounding backfill and bedding material, but is not intended to be a completely air and watertight enclosure. Overlaps shall be secured by the use of 2-inch wide, 10 mil thick, polyethylene pressure sensitive tape.

Any of the three different methods for the installation of polyethylene encasement on pipe are acceptable as described in AWWA C105. Methods A and B are for use with polyethylene tubes and Method C is for use with polyethylene sheets.

All fittings shall be covered in the same manner as the pipe. Where appurtenances are odd-shaped, they shall be covered with flat sheet, wrapped with seams doubled over, and taped. Branch services and other extensions shall be provided for with an X-shaped cut in the film, then the ends taped in place. At connections with unwrapped pipe or service pipes, the polyethylene shall extend 3 feet onto the unwrapped pipe and securely taped in place. Repair any damage encasement prior to backfill using polyethylene sheet and adhesive tape to secure all edges of the repair sheet.

- G. Installation of Special Restrained Joints: Restrained-joint pipe and ductile-iron retainer glands shall be installed according to manufacturer's recommendations. Torque wrenches

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and recommended special tools shall be used during installation. One set of special tools required shall be provided to the Owner.

3.3 INSTALLATION PVC PLASTIC PIPE:

- A. Pipe Laying: Pipe shall be laid with bell end facing in the direction of laying. Deflection in the pipe joints will be allowed as long as it does not exceed manufacturer's recommendations. Whenever it is necessary to deflect pipe from a straight line, either in the vertical or horizontal plane, to avoid obstructions or to plumb valve operators, the pipe itself may be uniformly curved as shown in the table below.

<u>Pipe Size</u> <u>C1.150 (inch)</u>	<u>Offset in</u> <u>20 foot length</u>	<u>Approximate Radius</u> <u>of Curve (feet)</u>
8	10	250
10	8	300
12	6	400
16	5	600

Pipe deflection for curvature shall not be permitted at temperatures less than 32°F ambient temperature.

- B. The cutting of pipe for inserting valves, fittings, or closure pieces shall be done in a neat and workmanlike manner without damage to the pipe, and so as to leave a smooth end at right angles to the axis of the pipe. Bevel the end of the pipe with a beveling tool after the pipe is field cut. Place a clearly visible position mark at the correct distance from the end of the field cut pipe.
- C. Jointing the Pipe: The outside of the spigot and the inside of the bell shall be thoroughly wiped clean. Set the rubber ring in the bell with the marked edge facing toward the end of the bell. Lubricate the spigot end using a thin film of the manufacturer-supplied lubricant. Push the pipe spigot into the bell manually, with blocking and bar or with special jacks. Position the completed joint so that the mark on the pipe end is in line with the end of the bell. Pipe joint shall not be assembled using power or trenching equipment. Do not insert the spigot end of the pipe beyond the "home" mark. The mark shall be visible and in line with the bell end in every case. The Inspector may require removal and reinsertion of the joint to the correct position at any time the mark is not visible after insertion. No exceptions to this requirement are allowed under any circumstances.
- D. Installation of Polyethylene Encasement: All metallic fittings and metallic mechanical restraints shall be polyethylene encased to prevent contact with the surrounding backfill and bedding material, but is not intended to be a completely air and watertight enclosure. Overlaps shall be secured by the use of multiple passes of 2-inch wide, 10 mil thick, polyethylene pressure sensitive tape. Polyethylene encasement shall be in accordance with AWWA C105

Where metallic appurtenances are odd-shaped, they shall be covered with flat sheet, wrapped with seams doubled over, and taped. Branch services and other extensions shall be provided for with an X-shaped cut in the film, then the ends taped in place. At connections with unwrapped pipe or service pipes, the polyethylene shall extend 3 feet onto the unwrapped pipe and securely taped in place. Repair any damage encasement

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prior to backfill using polyethylene sheet and adhesive tape to secure all edges of the repair sheet.

- E. Storage : PVC pipe shall not ever be stored outside in direct sunlight. Cover or store inside.

3.4 INSTALLATION OF HIGH DENSITY POLYETHYLENE (HDPE) – DIPS PIPE: High Density Polyethylene (HDPE) Pipe shall be installed in accordance with the instruction of the manufacturer, as shown on the Drawings and as specified herein. A factory qualified joining technician as designated by the pipe manufacturer shall perform all heat fusion joints.

- A. HDPE shall be installed either by Open Trench Construction or Directional Bore Method. Care shall be taken in loading, transporting and unloading to prevent damage to the pipe. Pipe or fitting shall not be dropped. All pipe or fitting shall be examined before installation, and no piece shall be installed which is found to be defective. Any damage to the pipe shall be repaired. If any defective pipe is discovered after it has been installed, it shall be removed and replaced with a sound pipe in a satisfactory manner by the Contractor.
- B. Under no circumstances shall the pipe or accessories be dropped into the trench or forced through a directional bore upon “pull-back”.
- C. Care shall be taken during transportation of the pipe such that it will not be cut, kinked, or otherwise damaged.
- D. Ropes, fabric or rubber protected slings and straps shall be used when handling pipes. Chains, cables or hooks inserted into the pipe ends shall not be used. Two slings spread apart shall be used for lifting each length of pipe.
- E. HDPE pipe shall not ever be stored outside in direct sunlight. Cover all pipe. Pipes shall be stored on level ground, preferably turf or sand, free of sharp objects, which could damage the pipe. Stacking of the polyethylene pipe shall be limited to a height that will not cause excessive deformation of the bottom layers of pipes under anticipated for project temperature conditions. Where necessary due to ground conditions, the pipe shall be stored on wooden sleepers, spaced suitably and of such width as not to allow deformation of the pipe at the point of contact with the sleeper or between supports.
- F. Pipe shall be stored on clean level ground to prevent undue scratching or gouging. The handling of the pipe shall be in such a manner that the pipe is not damaged by dragging it over sharp and cutting objects. The maximum allowable depth of cuts, scratches or gouges on the exterior of the pipe is 5 percent of wall thickness. The interior pipe surface shall be free of cuts, gouges, or scratches.
- G. Pipe shall be laid to lines and grade shown on the Drawings with bedding and backfill as shown on the Drawings.
- H. When laying is not in progress, including lunchtime, the open ends of the pipe shall be closed by fabricated plugs, or by other approved means.
- I. Sections of pipe with cuts, scratches or gouges exceeding 5 percent of the pipe wall thickness shall be removed completely and the ends of the pipeline rejoined.

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- J. The pipe shall be joined by the method of thermal butt fusion. All joints shall be made in strict compliance with the manufacturer's recommendations.
- K. Mechanical connections of the polyethylene pipe to auxiliary equipment such as valves, pumps and tanks shall be through flanged connections which shall consists of the following:
 - 1. A polyethylene flange shall be thermally butt-fused to the stub end of the pipe.
 - 2. A 316 stainless steel back up ring shall mate with a 316 stainless steel flange.
 - 3. A 316 stainless steel bolts and nuts shall be used.
- L. Flange connections shall be provided with a full-face neoprene gasket.
- M. HDPE pipe must be at the temperature of the surrounding soil at the time of backfilling and compaction.
- N. If a defective pipe is discovered after it has been installed, it shall be removed and replaced with a sound pipe in a satisfactory manner at no additional cost to the Owner. All pipe and fittings shall be thoroughly cleaned before installation, shall be kept clean until they are used in the work and when laid, shall conform to the lines and grades required, and shall not be kinked or otherwise damaged.
- O. Open Trench Installation:
 - 1. Section 31 23 33 shall apply in its entirety.
 - 2. The centerline of the pipe shall not deviate from a straight line drawn between the centers of the openings at the ends of the pipe by more than 1/16-in per foot of length. If a piece of pipe fails to meet this requirement check for straightness, it shall be rejected and removed from the site. Laying instructions of the manufacturer shall be explicitly followed.
 - 3. Alignment shall be preserved during installation. Deflection of the pipe shall occur only at those places on design drawings.
 - 4. Each length of the pipe shall have the assembly mark aligned with the pipe previously laid and held securely until enough backfill has been placed to hold the pipe in place. Joints shall not be "pulled" or "cramped".
 - 5. Precautions shall be taken to prevent flotation of the pipe in the trench.
 - 6. When moveable trench bracing such as trench boxes, moveable sheeting, shoring or plates are used to support the sides of the trench, care shall be taken in placing and moving the boxes or supporting bracing to prevent movement of the pipe, or disturbance of the pipe bedding and the backfill. Trench boxes, moveable sheeting, shoring or plates shall not be allowed to extend below top of the pipe. As trench boxes, moveable sheeting, shoring or plates are moved, pipe bedding shall be placed to fill any voids created and the backfill shall be recompacted to provide uniform side support for the pipe.

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7. Restrained joints shall be installed where shown on the Drawings.

3.5 SEWER SERVICE PIPELINE: When new sewer service pipelines or reconnection of existing services are to be installed, these pipelines are to be installed in accordance with the details set forth on the Drawings and with appropriate installation requirements.

The approximate location of the service lines are shown on the Drawings. Actual locations of these pipeline are to be verified in the field by the Contractor.

The Contractor shall tap and install a service saddle to new sewer pipelines only at locations to connect to existing mains not installed as part of this project. Connections onto reinforced concrete, ductile-iron, and PVC pipelines shall be made only by boring or drilling with equipment designed for this purpose. Connections shall not be made by impact equipment.

The Contractor shall remove from the pipeline all debris created by making connections before the service line is connected.

Service line saddle connections shall be attached to the main line with an epoxy bonding agent. The bonding agent shall be applied to a clean, dry surface. The connection shall remain dry until the bonding material has set, depending upon environmental conditions. Backfill around the connection shall not be attempted until the material has hardened. The Contractor shall repair damage to the main pipeline lining after the connecting PVC saddle has been bonded to the pipe. The lining shall be repaired with materials furnished by the manufacturer of the main pipeline.

The Contractor shall connect new service pipelines to fitting or saddle with the same material as the existing. Flexible couplings shall be used only to connect dissimilar piping materials or reconnecting existing services.

3.6 WATER SERVICES: Type K copper shall be bended in a manner that does not crimp or flatten the pipe. Provide insulators (insulating, dielectric unions) at all connections to ferrous piping.

3.7 EXTRA DEPTH EXCAVATION: In order to facilitate crossing under existing pipelines and other utilities, or as shown on the Drawings, the Contractor shall increase the depth of burial of new pipelines beyond design depth, as required.

3.8 INSTALLATION OF THRUST RESTRAINT: The movement of fittings shall be restrained by use of concrete thrust blocks and mechanical restraints.

Mechanical restraints shall be provided on joints located within or exceeding the lengths adjacent to fittings and valves specified on the detail on the Drawings. Install in strict conformance with the manufacturer's written instructions and recommendations. Measure and mark said limits of restraints adjacent to the trench with spray paint while the laying operation is underway.

Thrust blocks shall be poured between undisturbed solid ground and the fitting to be anchored; the area of bearing on the undisturbed trench wall shall be that shown on the thrust block detail. The bearing surface shall be neatly shaped: a flat vertical plane oriented perpendicular to the fitting. The concrete shall be so placed that the pipe or fitting joints will be accessible for repair. A bond breaker shall be placed over the fitting before placing concrete.

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Tie Rods: When specifically called for on the Drawings, install tie rods in addition to mechanical restraints and thrust blocks. Tie rods shall be assembled using clamps on each side of pipe bells with tie rods extending full pipe length for the dimensions shown on the detail drawings each direction from the restrained fitting. Clamp shall be installed tight enough to prevent twisting around the pipe. Two tie bolts per clamp with washer at clamp shall be located on side of pipe. Tighten tie bolt nut to “hand-tight” with 12-inch wrench (approximately 50-100 foot - pounds torque). Threaded tie rod ends shall extend two full threads past nut in final position.

3.9 WARNING TAPE: Install buried utility warning tape as shown on the trench detail shown on the Drawings, or centered 24-inches above the pipe within the trench backfill if not shown. Tape shall be laid continuously along the length of the pipeline.

3.10 FLUSHING, TESTING, AND DISINFECTING:

A. **Pipeline Flushing:** The Contractor shall flush the pipelines as the work progresses in accordance with good practice to ensure that sand, rocks or other foreign material are not left in any of the pipelines. If possible, the flushing shall be made through an open pipe end; otherwise, use of a fire hydrant may be acceptable if accessible and acceptable to the Owner.

B. **Alignment and Grade:** Pipelines conveying flow with gravity will be checked to determine whether any displacement of the pipe has occurred after the trench has been bedded to an elevation 6 inches above the pipe and tamped as specified. The test will be as follows:

A light will be flashed by means of a flashlight or by reflecting sunlight with a mirror into the pipe. If the illuminated interior of the pipeline shows poor alignment, displaced pipe, earth, or other debris in the pipe, or other kinds of defects, the defects, shall be remedied by the Contractor. The test will be repeated following completion of backfilling and poor alignment, displaced pipe, or other defects shall be corrected.

Curved sewers shall be cleaned with a heavy rubber ball after the pipe has been laid and the backfill has been placed. The rubber ball shall be similar and equal to a “Mac Wane Ball,” as manufactured by the Sidu Company, Long Beach, California. The ball shall be inflated and the outside diameter shall be the same as the inside diameter of the sewer pipe. The ball shall be placed at the upper manhole of the sewer reach to be cleaned. A head of water shall be placed against the ball to force the ball through the sewer, and debris flushed from the line shall be collected at the lower manhole, or its location. The ball shall be secured by a line to control the rate at which the ball is allowed to pass through the sewer.

C. **Deflection:** All pipelines conveying flow with gravity shall be tested for vertical deflection after placement and compaction of backfill. Method of testing shall be by deflectometer of the rigid GO/No-GO type device. Maximum allowable deflection shall be five percent (5%) of the pipe diameter. Pipe with vertical deflection greater than the allowable shall be excavated, removed from the pipeline, replaced, backfilled and compacted as specified, and retested.

D. **Pressure Test:** After each section of pipeline has been laid and partially backfilled, the pipe shall be slowly filled with water and tested. Pipe shall be tested at a minimum pressure of 100 psi at the lowest point in each section. Each section shall be tested

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separately, but transmission mains may be tested in convenient lengths. The duration of each pressure test shall be at least one continuous hour. Water used in testing the pipelines shall be taken from a potable water supply.

Each section of pipeline tested shall be slowly filled with water and all air removed. The specified test pressure shall be applied by means of a pump connected to the pipe. The Contractor shall furnish necessary labor, equipment, gauges, connection corporation stops to the pipeline, and other necessary appurtenances to perform the test.

No testing shall be permitted against valves or fittings that are a part of the existing system.

Exposed pipes, fittings, valves, hydrants, and joints will be carefully examined during the test. Cracked or defective pipe, fittings, valves or hydrants discovered during the pressure test shall be removed and replaced by the Contractor. The test shall be repeated until it is satisfactory.

Provide a minimum of 48 hours notice before the proposed test time. The Contractor may complete their own preliminary tests prior to the witnessed test to confirm readiness and that all air has been purged from the pipeline which may interfere with the test results.

Reporting: The Contractor shall prepare and submit a written statement or form documenting each pressure test which shall include at a minimum the following:

1. Description and identification of piping tested (location, station-to-station, etc).
2. Test pressure, beginning and ending values
3. Date and beginning and ending times of the test.
4. Witnessing by Contractor and Engineer – names and signatures.
5. Test evaluation. Note pass or fail.
6. Remarks, to include such items as:
 - a. Leaks (type, location).
 - b. Repairs made on leaks.

- E. Leakage Test: If the pipeline does not pass the pressure test, meaning there was a measured drop in pressure during the pressure test period, then a leakage test shall be conducted. The Contractor shall furnish the pump, pipe, gauges, connections, flow meters, and other necessary apparatus, and shall furnish all necessary assistance to conduct the test. The duration of each leakage test shall be two hours, and, during the test, the main shall be subjected to a hydrostatic pressure of 150 psi.

No pipeline installation will be acceptable until the leakage is less than the amount computed by the following formula:

$$1. \quad \text{DIP, PVC} \quad L = \frac{SD(P)^{0.5}}{133,200}$$

L = Allowable leakage in gallons (per hour)

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S = Tested length of pipe (feet)
D = Nominal diameter of pipe (inches)
P = Average test pressure during the test (psi)

2. HDPE Allowable leakage is none.

Should a test of pipe installed disclose leakage greater than that specified above, the Contractor shall, locate and repair the points of leakage until the leakage is within the specified allowance.

The pipe may be subjected to hydrostatic pressure, inspected, and tested for leakage at a convenient time after the trench has been partially backfilled, except at the joints. Where a section is provided with concrete thrust blocks, the pressure test shall not be made until at least two days have elapsed after the concrete was installed.

Contractor shall provide 48 hours notification before the pipe is to be tested.

F. Testing of Copper Services: All copper service lines, corporation stops, insulators, curb stops, fittings, and splices shall be leak tested under pressure prior to acceptance and placing into service. Contractor shall facilitate testing by pressurizing the service (either by opening the corp stop at the main or by temporary pump) and by providing shoring and safe access to the trench. Allowable leakage shall be none.

- END OF SECTION -